Ex: Newtonian mechanics: we have a force F acting on a system $m\frac{d^2x}{dt^2} = F$

if $\nabla \times F = 0$ then F is conservative $\rightarrow F = -\nabla V$ V potential

if $V \rightarrow V + c$ $c \in \mathbb{R}$ $F \rightarrow F$

Why can we do Lorenz gauge?

A, ϕ arbitrary let $A' = A + \nabla\lambda$ $\phi' = \phi - \frac{\partial\lambda}{\partial t}$

can we choose λ so that $\nabla \cdot A' + \frac{\partial \phi'}{\partial t} = 0$?

$$\nabla \cdot A' + \frac{\partial \phi'}{\partial t} = \nabla \cdot A + \nabla^2\lambda + \frac{\partial\phi}{\partial t} - \frac{\partial^2\lambda}{\partial t^2} \rightarrow \boxed{\frac{\partial^2\lambda}{\partial t^2} - \nabla^2\lambda = \overbrace{\nabla \cdot A + \frac{\partial\phi}{\partial t}}^{\text{fixed source}}}$$

wave eq with source for λ

Relativistically 4-position: (t, x, y, z)

4-potential $(A_\mu) = (-\phi, \vec{A})$ $A = A_\mu dx^\mu$ (1-form)

gauge transformation: $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$ || Lorenz gauge: $\partial^\mu A_\mu = 0$

$\hookrightarrow \eta^{\mu\nu}\partial_\mu A_\nu = 0$

$= \partial_0 A_0 - \partial_1 A_1 - \partial_2 A_2 - \partial_3 A_3 = 0$

$= -\frac{\partial\phi}{\partial t} - \nabla \cdot A = 0$

Faraday tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ || $F_{\mu\nu} = -F_{\nu\mu}$

F is a 2-form

often in relativistic physics

greek indices $\mu = 0, 1, 2, 3$

latin indices $i = 1, 2, 3$

$A_0 = -\phi$

$$F_{0i} = \partial_t A_i + \partial_i \phi = -E_i$$

$i = 1, 2, 3$

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

$$F_{ij} = \partial_i A_j - \partial_j A_i \qquad B = (\nabla \times A) = \begin{pmatrix} \partial_2 A_3 - \partial_3 A_2 \\ \partial_3 A_1 - \partial_1 A_3 \\ \underbrace{\partial_1 A_2 - \partial_2 A_1}_{B_z} \end{pmatrix} \rightarrow \begin{matrix} F_{12} = B_z & F_{31} = B_y & F_{23} = B_x \\ F_{21} = -B_z & F_{13} = -B_y & F_{32} = -B_x \end{matrix}$$

$$(F_{\mu\nu}) = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} \rightarrow \text{thing we can measure}$$

$$\mathcal{L} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial_\alpha A_\beta - \partial_\beta A_\alpha)$$ free Lagrangian for photons

$\hookrightarrow \frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}$

$$\mathcal{L} = \mathcal{L}(A, \partial A) = \frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial_\alpha A_\beta - \partial_\beta A_\alpha)$$

$$= \frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta}\left[\underline{\partial_\mu A_\nu \partial_\alpha A_\beta} - \partial_\mu A_\nu \partial_\beta A_\alpha - \partial_\nu A_\mu \partial_\alpha A_\beta + \underline{\partial_\nu A_\mu \partial_\beta A_\alpha}\right]$$

$\mu \leftrightarrow \beta$

$\nu \leftrightarrow \alpha$

$$\eta^{\mu\alpha}\eta^{\nu\beta}\,\partial_\mu A_\nu \partial_\alpha A_\beta$$
$$\updownarrow$$
$$\eta^{\beta\nu}\eta^{\alpha\mu}\,\partial_\alpha A_\beta \partial_\mu A_\nu$$

$$= \frac{1}{2}\eta^{\rho\alpha}\eta^{\sigma\beta}\left(\underline{\partial_\rho A_\sigma}\,\partial_\alpha A_\beta - \underline{\partial_\rho A_\sigma}\,\partial_\beta A_\alpha\right)$$

<u>EL eqs</u> ? $\quad \dfrac{\partial \mathcal{L}}{\partial A_\mu} = 0 \Rightarrow \boxed{\dfrac{\partial}{\partial x^\mu}\dfrac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} = 0}$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = \frac{1}{2}\eta^{\mu\alpha}\eta^{\nu\beta}(\partial_\alpha A_\beta - \partial_\beta A_\alpha) + \frac{1}{2}\eta^{\rho\mu}\eta^{\sigma\nu}\partial_\rho A_\sigma - \frac{1}{2}\eta^{\rho\nu}\eta^{\sigma\mu}\partial_\rho A_\sigma$$

$\updownarrow \rho \to \alpha,\ \sigma \to \beta$ $\qquad\qquad$ $\downarrow \rho \to \beta,\ \sigma \to \alpha$

$$= \frac{1}{2}\eta^{\mu\alpha}\eta^{\nu\beta}(\partial_\alpha A_\beta - \partial_\beta A_\alpha) + \frac{1}{2}\eta^{\mu\alpha}\eta^{\nu\beta}\partial_\alpha A_\beta - \frac{1}{2}\eta^{\mu\alpha}\eta^{\nu\beta}\partial_\beta A_\alpha$$

$$= \eta^{\mu\alpha}\eta^{\nu\beta}(\partial_\alpha A_\beta - \partial_\beta A_\alpha) = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta} = F^{\mu\nu}$$

$\rightarrow$ <u>EL</u>: $\boxed{\partial_\mu F^{\mu\nu} = 0}$

$$\eta = \mathrm{diag}(+1, -1, -1, -1)$$

$$(F_{\mu\nu}) = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}$$

$$F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\mu\beta}F_{\alpha\beta}$$

$$F^{0i} = E_i$$

$$F^{ij} = F_{ij}$$

$$(F^{\mu\nu}) = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}$$

$$\varepsilon_{ijk}F^{ij} = B_k$$

$$\rightarrow = \begin{cases} +1 & \text{if } (ijk) \text{ even permutation of } (123) \\ -1 & \text{if } (ijk) \text{ odd permutation} \\ 0 & \text{if any indices are repeated} \end{cases}$$

$$\partial_\mu F^{\mu i} = 0 \rightarrow \partial_0 F^{0i} + \partial_1 F^{1i} + \partial_2 F^{2i} + \partial_3 F^{3i} = 0$$

$$(\nabla \times A) = \begin{pmatrix} \partial_2 A_3 - \partial_3 A_2 \\ \partial_3 A_1 - \partial_1 A_3 \\ \partial_1 A_2 - \partial_2 A_1 \end{pmatrix}$$

$$(\nabla \times A)_k = \sum_{i,j} \varepsilon_{ijk} \partial_i A_j$$

$$(\nabla \times A)_3 = \sum_{i,j=1,2} \varepsilon_{ijk} \partial_i A_j$$

$$= \varepsilon_{123} \partial_1 A_2 + \varepsilon_{213} \partial_2 A_1$$

$$= \partial_1 A_2 - \partial_2 A_1$$

$$\frac{1}{2} \varepsilon_{ijk} F^{ij} = B_k$$

$$F_{12} = B_z \quad F_{31} = B_y \quad F_{23} = B_x$$

$$F_{21} = -B_z \quad F_{13} = -B_y \quad F_{32} = -B_x$$

$$B_3 = F^{12} = \varepsilon_{123} F^{12}$$

$$= -F^{21} = \varepsilon_{213} F^{21}$$

$$B_3 = \frac{1}{2} \varepsilon_{ij3} F^{ij}$$

$$B_2 = F^{31} = \varepsilon_{312} F^{31}$$

$$= -F^{13} = \varepsilon_{132} F^{13}$$

$$\boxed{F^{ij} = \sum_k \varepsilon^{ijk} B^k}$$

$$\partial_\mu F^{\mu i} = 0 = \partial_0 F^{0i} + \partial_j F^{ji} = \partial_t E^i + \varepsilon^{jik} \partial_j B_k$$

$$= \partial_t E^i - \varepsilon^{ijk} \partial_j B_k \quad (\nabla \times B)_i$$

$$\Rightarrow (\nabla \times B - \partial_t E)_i = 0$$

$$\partial_\mu F^{\mu 0} = 0 = \cancel{\partial_0 F^{00}} + \partial_i \overbrace{F^{i0}}^{= -E^i} = 0 \quad \Rightarrow \quad \nabla \cdot E = 0$$

$$\mathcal{L} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

<u>Dirac equation</u> $\Rightarrow$ attempt at finding a relativistic equivalent of Shrodinger's eq.

<u>Shrod:</u> $$-i\hbar \underbrace{\frac{\partial}{\partial t}}_{1st} \psi + \frac{\hbar^2}{2m} \underbrace{\nabla^2}_{2nd} \psi = 0$$

Klein-Gordon $$\frac{\partial^2 \psi}{\partial t^2} - \nabla^2 \psi = 0$$

Dirac 1st order in time and space

$$\underbrace{(i\hbar\gamma^\mu\partial_\mu - m\mathbb{1})}_{\text{matrix}}\underbrace{\psi}_{\text{column vector}} = 0$$

ψ not a scalar field anymore

γ^μ are 4 matrices

usually ψ has 4 components and γ^μ are 4×4 matrices

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types of eqs:
- Klein-Gordon: scalar eq (φ scalar field)
- Electromagnetism: vector eq (A_μ vector field)
- Dirac: spinor eq (ψ is a spinor)

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if ψ satisfies Dirac eq. then it also satisfies Klein-Gordon

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}$$

$\{\gamma^\mu,\gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu$ anti-commutator

$\rightarrow$ Clifford algebra

different representations of γ matrices

standard rep. $\gamma^0 = \begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix}$ $\gamma^1 = \begin{pmatrix} 0 & \sigma_x \\ -\sigma_x & 0 \end{pmatrix}$ $\gamma^2 = \begin{pmatrix} 0 & \sigma_y \\ -\sigma_y & 0 \end{pmatrix}$ $\gamma^3 = \begin{pmatrix} 0 & \sigma_z \\ -\sigma_z & 0 \end{pmatrix}$

imaginary

Lagrangian for Dirac

$\bar\psi\gamma^\mu\overleftrightarrow{\partial_\mu}\psi$

$\bar\psi = \psi^*\gamma^0$

$$\mathcal{L} = i\hbar\frac{1}{2}\left(\bar\psi\gamma^\mu\partial_\mu\psi - \partial_\mu\bar\psi\gamma^\mu\psi\right) - m\bar\psi\psi$$

ψ and $\bar\psi$ are treated as independent fields

ψ is complex

ψ and its complex conjugate are independent as real variables

if $z = x+iy$ $\quad x = \frac{z+\bar z}{2}$

$\bar z = x - iy$ $\quad y = \frac{z-\bar z}{2i}$

we could do something like this: $f: \mathbb{C}\to\mathbb{R}$ or think of it as $f:\mathbb{R}^2\to\mathbb{R}$

$\frac{\partial}{\partial x}f$, $\frac{\partial}{\partial y}f$ make sense

$\frac{\partial}{\partial z}f = ?$ $\quad \frac{\partial}{\partial \bar z}f = ?$

$z = x+iy$ $\quad x = \frac{z+\bar z}{2}$

$$\frac{\partial}{\partial z} = \frac{\partial x}{\partial z}\frac{\partial}{\partial x} + \frac{\partial y}{\partial z}\frac{\partial}{\partial y} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

$$\frac{\partial}{\partial \bar z} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

$$\boxed{\mathcal{L} = i\hbar \tfrac{1}{2}\left(\bar{\Psi}\gamma^\mu \partial_\mu \Psi - \partial_\mu \bar{\Psi}\gamma^\mu \Psi\right) - m\bar{\Psi}\Psi}$$

Free fermion with mass m

$$\frac{\partial \mathcal{L}}{\partial \bar{\Psi}} = \frac{i\hbar}{2}\gamma^\mu \partial_\mu \Psi - m\Psi$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \bar{\Psi}} = -i\hbar \frac{1}{2}\gamma^\mu \Psi$$

$$\rightarrow \frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}}{\partial \partial_\mu \bar{\Psi}} = \frac{\partial \mathcal{L}}{\partial \bar{\Psi}}$$

$$-\frac{i\hbar}{2}\gamma^\mu \partial_\mu \Psi = \frac{i\hbar}{2}\gamma^\mu \partial_\mu \Psi - m\Psi$$

$$\boxed{i\hbar\gamma^\mu \partial_\mu \Psi - m\Psi = 0}$$ Dirac eq.

$$\frac{\partial \mathcal{L}}{\partial \Psi} = -\frac{i\hbar}{2}\partial_\mu \bar{\Psi}\gamma^\mu - m\bar{\Psi}$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \bar{\Psi}} = -i\hbar\frac{1}{2}\bar{\Psi}\gamma^\mu$$

$$\rightarrow \boxed{i\hbar\partial_\mu \bar{\Psi}\gamma^\mu + m\bar{\Psi} = 0}$$ dual of Dirac eq.

$$(\gamma^0)^* = \gamma^0 \qquad (\gamma^k)^* = -\gamma^k$$

$$\gamma^0\gamma^k + \gamma^k\gamma^0 = 0 = 2\eta^{0k} \qquad \underbrace{\gamma^0\gamma^k = -\gamma^k\gamma^0}$$

$$0 = \left(i\hbar\partial_\mu \bar{\Psi}\gamma^\mu + m\bar{\Psi}\right)^*$$

$$= -i\hbar(\gamma^\mu)^*\partial_\mu \bar{\Psi}^* + m\bar{\Psi}^*$$

$$= -i\hbar\underbrace{(\gamma^\mu)^*\gamma^0}\partial_\mu \Psi + \gamma^0 m\Psi$$

$$\bar{\Psi} = \Psi^*\gamma^0$$

$$\bar{\Psi}^* = (\gamma^0)^*\Psi = \gamma^0\Psi$$

$$= -i\hbar\left[\gamma^0\gamma^0\partial_0\Psi + (\gamma^k)^*\gamma^0\partial_k\Psi\right] + \gamma^0 m\Psi$$

$$= -i\hbar\left[\gamma^0\gamma^0\partial_0\Psi - \gamma^k\gamma^0\partial_k\Psi\right] + \gamma^0 m\Psi$$

$$= -i\hbar\left[\gamma^0\gamma^0\partial_0\Psi + \gamma^0\gamma^k\partial_k\Psi\right] + \gamma^0 m\Psi$$

$$= \gamma^0\left[-i\hbar\left(\gamma^0\partial_0\Psi + \gamma^k\partial_k\Psi\right) + m\Psi\right] \Rightarrow -i\hbar\gamma^\mu\partial_\mu\Psi + m\Psi = 0$$

Putting together fields: if $\mathcal{L}_1(\Psi, \partial\Psi)$ is the free Lagrangian for fermionic field (Dirac) and $\mathcal{L}_2(A, \partial A)$ is free Lagrangian for photons

start by looking at $\mathcal{L}_{free} = \mathcal{L}_1 + \mathcal{L}_2$

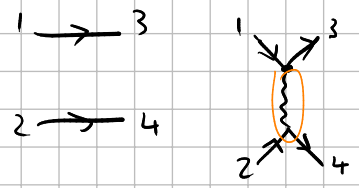
then we look at $\mathcal{L} = \mathcal{L}_{free} + \mathcal{L}_{int}$

interaction

$$\mathcal{L}_{Fermion} \rightarrow i\hbar\frac{1}{2}\left(\bar{\Psi}\gamma^\mu\underbrace{(\partial_\mu - ieA_\mu)}\Psi - (\partial_\mu + ieA_\mu)\bar{\Psi}\gamma^\mu\Psi\right) - m\bar{\Psi}\Psi$$

$\rightarrow$ covariant derivative

$$\mathcal{L}_{int} = \hbar e\,\bar{\Psi}\gamma^\mu A_\mu \Psi$$



virtual photon

$$\mathcal{L} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{i\hbar}{2}\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi + \boxed{\hbar e\,\bar{\Psi}\gamma^\mu A_\mu\Psi} + c.c.$$

Quantize things

- QM: x, p parameters describing system
  (position, momentum)
  make them into operators $\hat{x}, \hat{p}$ acting on Hilbert space

- second quantization: use t, x, y, z as spacetime parameters.
  take fields $\varphi, \Psi, A, \ldots$ and make $\varphi(x)$ into an operator
  $\hat{\varphi}$ operator-valued field (canonical quantization)

  in QFT we tend to use path integrals

$\mathcal{L}$ lagrangian describing fields and interactions

build action $S[\varphi, \Psi, A \ldots] = \int_M d^4x\, \mathcal{L}(\varphi(x), \partial\varphi(x), \ldots)$ number

path integral: sum over all possible fields with a weight

$$\boxed{\int D\varphi\, e^{\frac{i}{\hbar}S[\varphi]}}$$

Why path integral?

$q(t)$ path in physical space

in QM. $H = \frac{p^2}{2m} + V(q)$

canonical

$q \to \hat{q}$

$p \to \hat{p}$

$H \to \hat{H}$

$\mathcal{H} = L^2(\mathbb{R}, \mathbb{C})$

$\hat{q}\,\psi(q) = q\,\psi(q)$

$\hat{p}\,\psi(q) = -i\hbar \frac{\partial}{\partial q}\psi(q)$

$\mathcal{L} = \frac{1}{2} m v^2 - V(q)$

$\downarrow$ path integral

$$\int Dq \; e^{\frac{i}{\hbar}\int dt\, \mathcal{L}(q,\dot{q})}$$

<u>Second quantization:</u> $\hat{\varphi}(x)|0\rangle$ = state with one particle at position x

probability amplitudes

$\langle 0| T(\hat{\varphi}(x)\hat{\varphi}(y))|0\rangle$ = 2 point correlation function

$\to$ propagator

$$\boxed{\langle T(\hat{\varphi}(x_1)\cdots\hat{\varphi}(x_n))\rangle = \frac{1}{\int D\varphi\, e^{\frac{i}{\hbar}S[\varphi]}} \int D\varphi\; e^{\frac{i}{\hbar}S[\varphi]}\, \varphi(x_1)\cdots\varphi(x_n)}$$

ex: Gaussian $\langle x^2\rangle = \frac{1}{\int dx\, e^{-ax^2}} \int dx\; e^{-ax^2}\, x^2$

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① <u>free fields</u> 1 scalar massless boson

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi = \frac{1}{2}\left[(\partial_t\phi)^2 - (\nabla\phi)^2\right]$$

Wick rotation $t \to -it$ $\quad \partial_t \to i\partial_t \quad \to \mathcal{L} = -\frac{1}{2}\left((\partial_t\phi)^2 + (\nabla\phi)^2\right)$

$$i\int d^4x = i\int dt\,dx\,dy\,dz \to \int dt\,dx\,dy\,dz$$

by analogy we come up with a way to assign a value to $\int D\varphi\; e^{-\tilde{S}[\varphi]}$

Gaussian integral: $\int dx^n\; e^{-x^t A x}$ $\quad$ A $n\times n$ positive definite matrix

$$= \sqrt{\det(2\pi A^{-1})}$$

$$\int dx^n\; e^{-x^t A x}\, x_i x_j = \sqrt{\det(2\pi A^{-1})}\; (A^{-1})_{ij}$$

propagator

n degrees of freedom $\Rightarrow$ infinite (continuous) d.o.f

discrete field $(\phi_1, \phi_2, \dots, \phi_n)$ || continuous field ϕ_x x continuous

$(\varphi : \{1, 2, \dots, n\} \to \mathbb{R})$

$$\int dt \; \partial_t \phi \, \partial_t \phi = \cancel{\text{boundary term}}^{\to 0} - \int dt \; \partial_t^2 \phi \, \phi$$

$$\int d^4x \; (\partial_t \phi)^2 + (\nabla \phi)^2 = - \int d^4x \; \phi \overbrace{(\partial_t^2 + \nabla^2)}^{\text{positive-def matrix}} \phi$$

by analogy: $$\int D\varphi \; e^{-\int d^4x \, \phi (\partial_t^2 + \nabla^2) \phi} \; \phi(x) \phi(y) \propto (\partial_t^2 + \nabla^2)^{-1}(x, y)$$

propagator/Green's function

$$\langle T(\hat{\varphi}(x) \hat{\varphi}(y)) \rangle = \frac{1}{\int D\varphi \, e^{-S \dots}} \int D\varphi \; e^{-\int d^4x \, \phi (\partial_t^2 + \nabla^2) \phi} \; \phi(x) \phi(y) = (\partial_t^2 + \nabla^2)^{-1}(x, y)$$

$$\downarrow$$

$$(p^2)^{-1} = \frac{1}{p^2}$$

<u>Fourier transforms</u> $x \to p$ $\quad \mathcal{F}(f'(x)) = p \, \mathcal{F}(f)(p)$

add mass term: $$\int d^4p \frac{1}{p^2 + m^2 + i\varepsilon}$$

self-interaction

$$\int dx \; e^{-ax^2 + gx^4} \approx \int dx \; e^{-ax^2} (1 + gx^4)$$

φ^3 interaction